A065 v0.3.6 - Analytic generator and epsilon audit

BLIND; geometric and numerical diagnostic only. No Euler/Casimir/gravity closure.

Core smoothing

Regularized segment kernel: (distance^2 + epsilon^2)^(-3/2). epsilon=0.06 is a dimensionless smoothing scale in the original historical curve coordinate units; not a derived SST radius.

Gamma=1, T=0.04 and RK4 steps=8 fixed. Mesh refinement changes N, never epsilon.

Runtime-generated historical trefoils

Independent direct generator vs pinned originals: max coordinate discrepancy 3.494e-10

The original SciPy regenerate.py is preserved and hashed; the falsifier uses equivalent stdlib arc-length inversion with independent comparison to sealed XYZ.

A_Fourier_2015_p32

N=600: mean ds=0.033293356, ds/eps=0.554889, edge CV=2.653e-05

N=1200: mean ds=0.016646996, ds/eps=0.27745, edge CV=6.633e-06

N=2400: mean ds=0.0083235376, ds/eps=0.138726, edge CV=1.658e-06

N=4800: mean ds=0.0041617738, ds/eps=0.0693629, edge CV=4.150e-07

600_vs_1200: relative error=0.849902 %, 1pct pass=True; 600vs1200 is a diagnostic, not fine-pair veto.

1200_vs_2400: relative error=0.211758 %, 1pct pass=True; 600vs1200 is a diagnostic, not fine-pair veto.

2400_vs_4800: relative error=0.0528703 %, 1pct pass=True; 600vs1200 is a diagnostic, not fine-pair veto.

B_Torus_2015_p66

N=600: mean ds=0.053162331, ds/eps=0.886039, edge CV=1.175e-05

N=1200: mean ds=0.026581917, ds/eps=0.443032, edge CV=2.938e-06

N=2400: mean ds=0.013291052, ds/eps=0.221518, edge CV=7.347e-07

N=4800: mean ds=0.0066455379, ds/eps=0.110759, edge CV=1.844e-07

600_vs_1200: relative error=2.65588 %, 1pct pass=False; 600vs1200 is a diagnostic, not fine-pair veto.

1200_vs_2400: relative error=0.679595 %, 1pct pass=True; 600vs1200 is a diagnostic, not fine-pair veto.

2400_vs_4800: relative error=0.169058 %, 1pct pass=True; 600vs1200 is a diagnostic, not fine-pair veto.

Falsifier gate policy

G19 = analytic regenerated vs pinned XYZ. G20 = dimensionless ds/epsilon diagnostic; 0.25 at N4800 only. G21 = native generated epsilon=0.06 1200->2400 and 2400->4800 spatial comparison under 1pct.

Do not infer physical r_c from epsilon. All historic FAIL statuses and reveal block remain.

prereg_sha256: 7ca11af33d2e48fdc3293b6153e76ce97be403b250cf54afe382fe89518a15bd